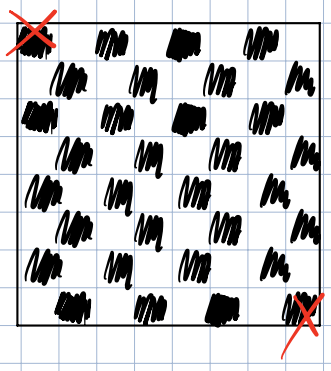
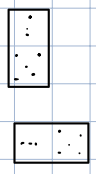
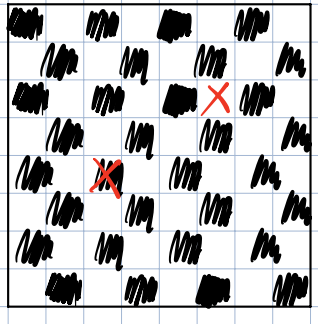


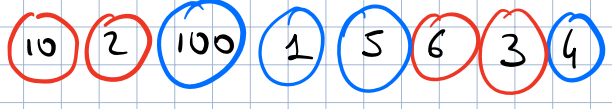


Is IT possible to cover this chess board with dominoes? NO!



Alice & Bob

2n numbers




A 10 2 3 6 = 21

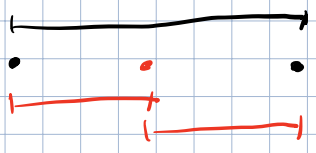
B 4 100 1 5 = 110

Is there any strategy for Alice to get always at least as Bob?

# Binary Search for the answer

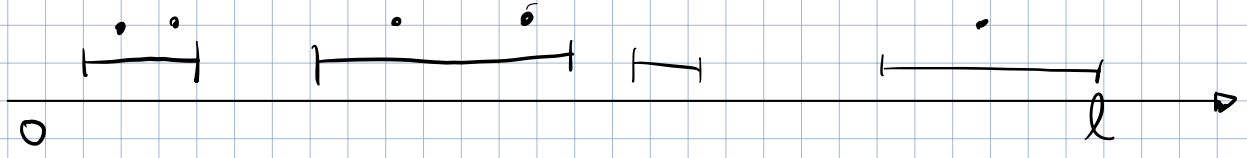
$\sqrt{x}$

## Social Distancing



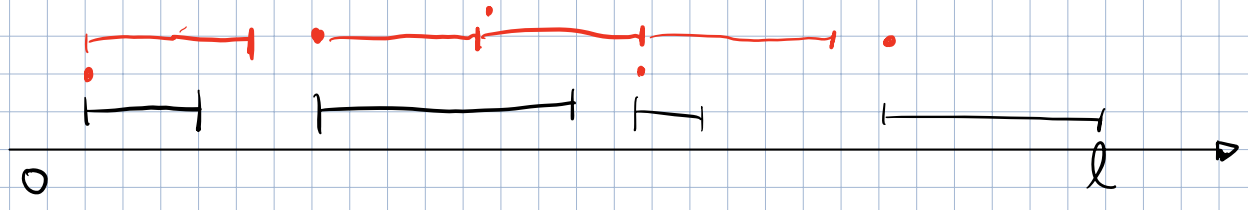
$n$ disjoint intervals

$c$ ($\leq 5$) people to place on these intervals




**Goal:** Find $c$ points on the intervals such that the smallest distance between any of them is maximized

Target distance $d = 10$



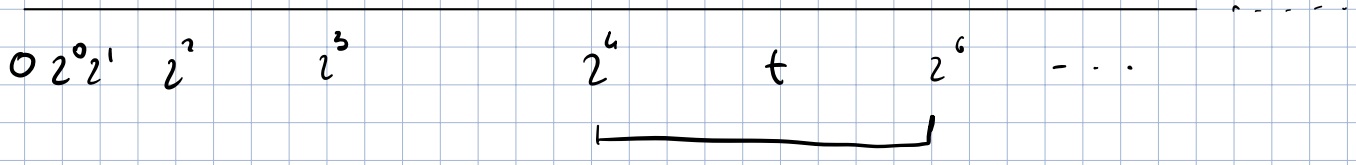

PLACE POINTS WITH TARGET DISTANCE

$\Theta(n+c)$ time

binary search for the answer $d^* \in [0, \frac{\ell}{c}]$ with $\log(\frac{\ell}{c})$ target distances to try in $\Theta(n+c)$ time each

FINAL COMPLEXITY IS $\Theta((n+c)\log\frac{\ell}{c})$ time

## EXPONENTIAL (BINARY) SEARCH

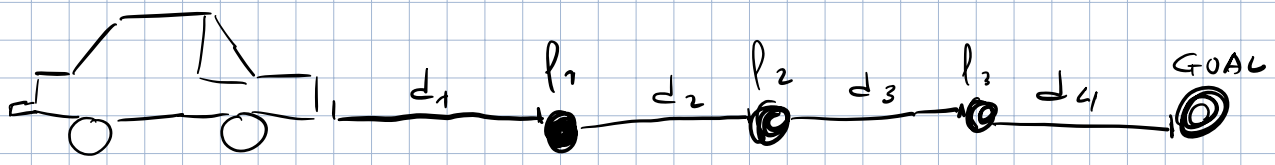



Phase #1 try powers of 2 till the first true

$\Theta(\log(t))$ time

Phase #2 binary search in $[2^i \dots 2^{i+1}]$

$\Theta(\log(t))$ time




$f$ = fuel/km

$G^*$ min amount of fuel in our tank?

# Two-pointers trick

## Problem

### Longest k-good segment

Given $A[1, n]$ of values and a value $k$

A subarray is called k-good if it contains no more than $k$ different values.

4 2 3 1 2 2 3 4

3-good

Find (a) longest one.




$k = 3$

Time: $\Theta(n(\log^* n))$ time

H

C = ✗ ✗ ✗ S

# Sweep Line

## Maximum Number of overlapping intervals

Given $n$ intervals $[s_i, e_i]$ where $e_i, s_i \in \mathbb{N}$

we say that two intervals overlap

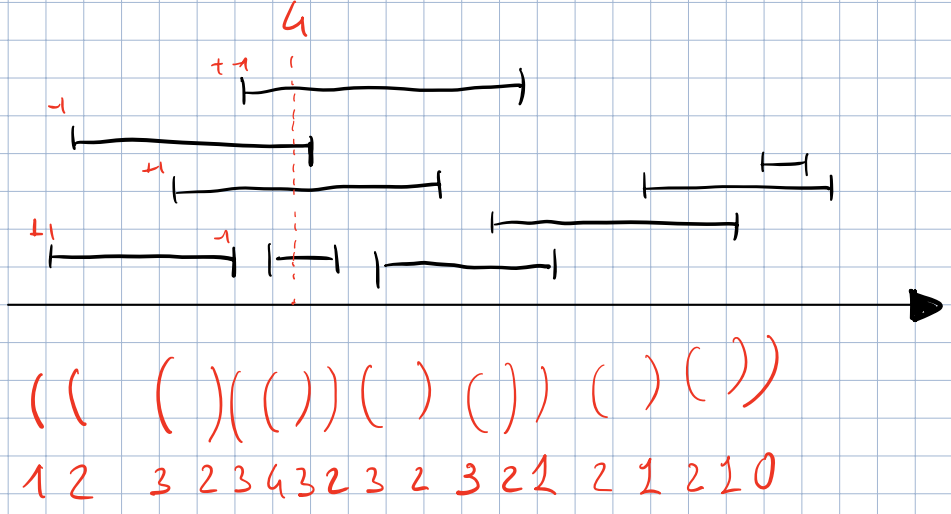
if $\exists x$ such that

$$x \in [s_i, e_i]$$

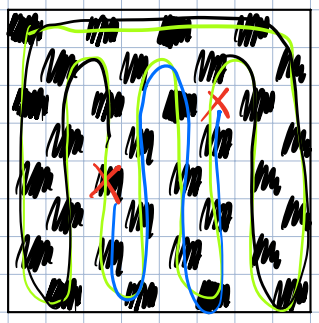
AND

$$x \in [s_j, e_j)$$

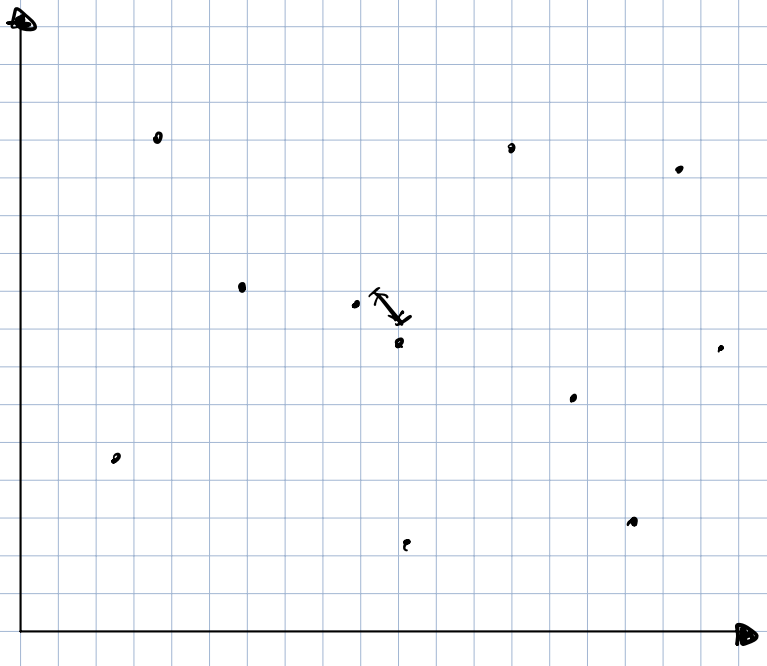
Compute the maximum number of overlapping intervals




Time complexity $\Theta(n \log n)$ time



# Closest Pair of Points




Find the closest pair of points

Euclidean Distance

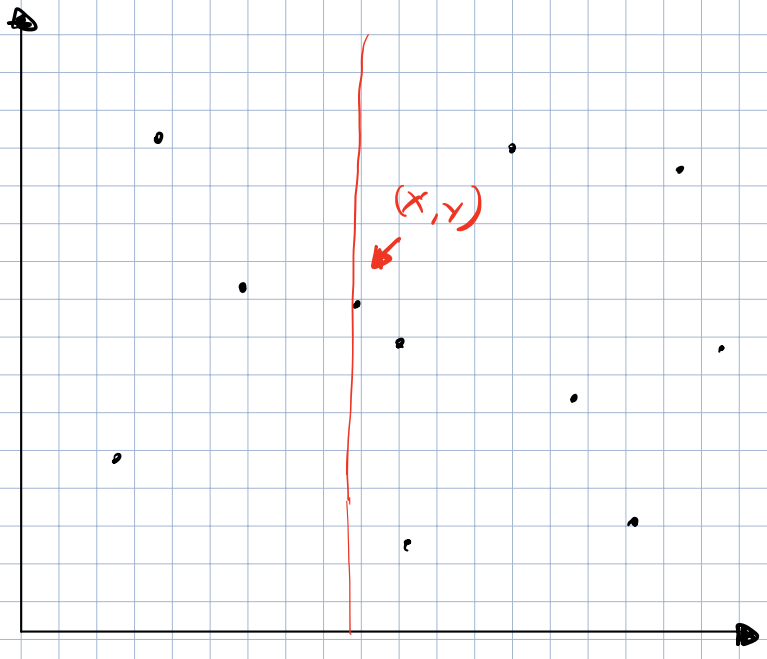
$$d((x_1, y_1), (x_2, y_2)) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

# Solution

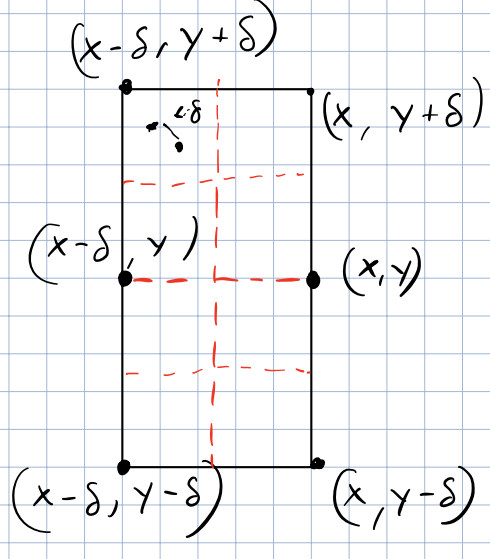
Brute-Force Solution $\Theta(n^2)$ time

Target Solution $\Theta(n \log n)$ time

$\delta$ = smallest distance seen so far

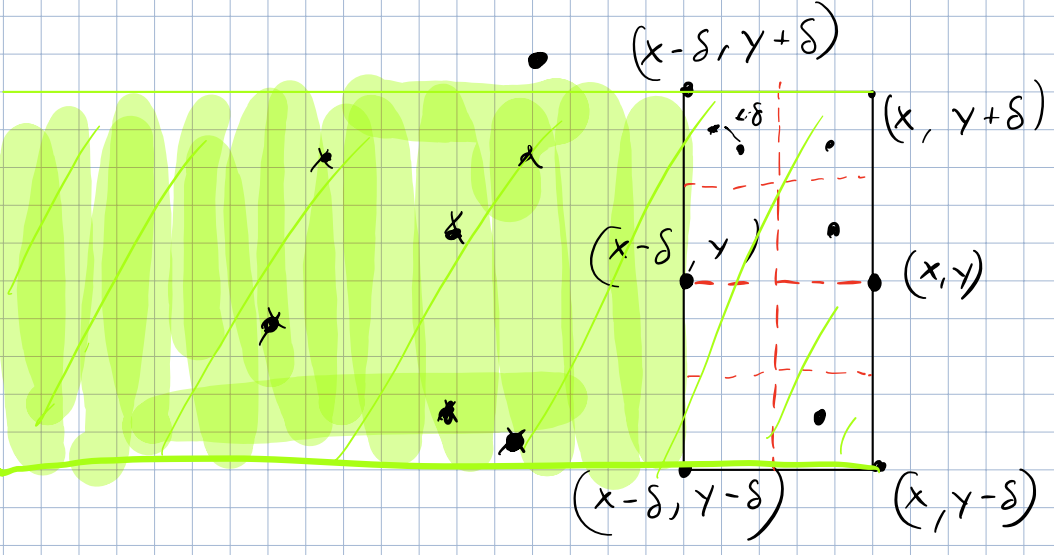



we know $\delta$




It is possible to prove that there are at most 6 points in the rectangle

- Store already processed points into a BST sorted by y

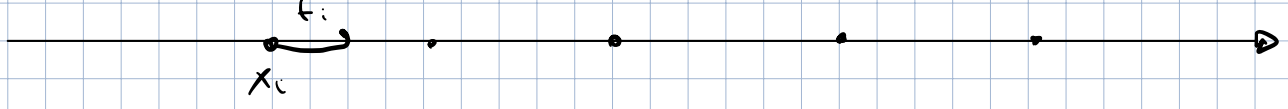



Time complexity : $\Theta(n \log n)$ time

# Problem

## Frogs and Mosquitos

There are n frogs on the x-axis



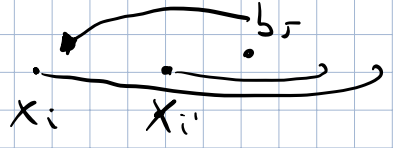

Each frog (i) has a value $t_i$ which is the length of its tongue

m mosquitos start landing on x-axis

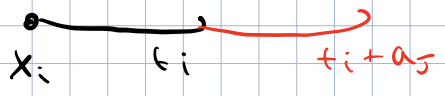
A mosquito j lands at position $b_j$ and it has a value $a_j$.

A frog i eats a mosquito $b_j$ if $b_j \in [x_i, x_i + t_i]$.

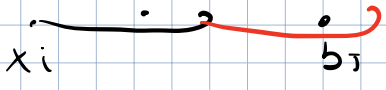
if there are many frogs able to eat $b_j$ only the left most one eats it.



If frog $x_i$ eats $b_j$ with value $a_j$ then the new tongue is $t_i + a_j$



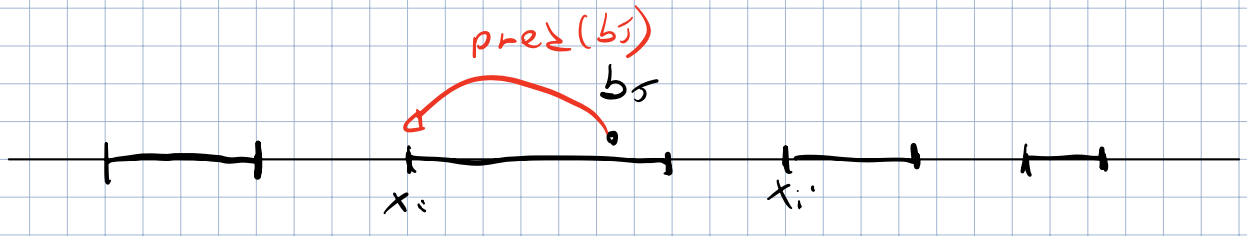

Notice that a mosquito $b_j$ may be not eaten for a while




Target time complexity $\Theta((n+m)\log(n+m))$

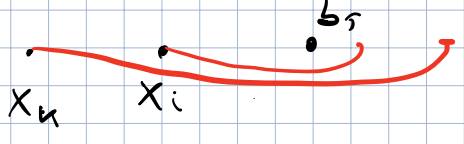
Dynamic 1D point location

Solution

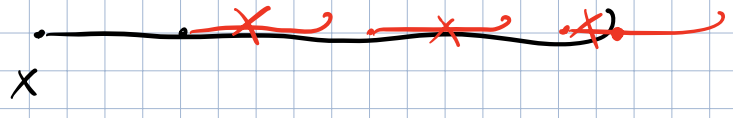



- Storing $x_i$ in a BST together with $t_i$
- Predecessor of $b_j$ is $x_i$ which is the frog that eats the mosquito
  - $\Theta(\log(n))$

- Issue #1




- Preprocess intervals to remove overlaps




- takes $\Theta(n \log n)$ time

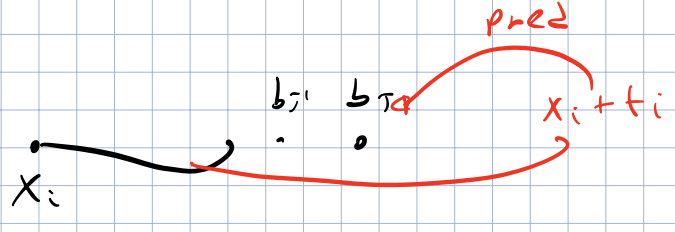
- Issue #2 : A interval is growing to righ. we are introducing new overlaps

repeat
- move to next prog
  - remove it if now total overlap
  - update its starting point

$\Theta(\log n)$ because just for the last one

overall cost is $\Theta(n \log n)$

- Issue #3 not eaten mosquitos
  - Store mosquitos in a BST
  - Check if the last prog can eat a old mosquito




$\Theta(m \log m)$

# Alice & Bob

2n numbers

Odd = 118
Even = 13

odd even odd even odd even odd even

10 2 100 1 5 6 3 4

A

B

Is there any strategy for Alice to get always at ast as Bob?